文献汇报

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2020.04.09

Towards High Performance Video Object Detection

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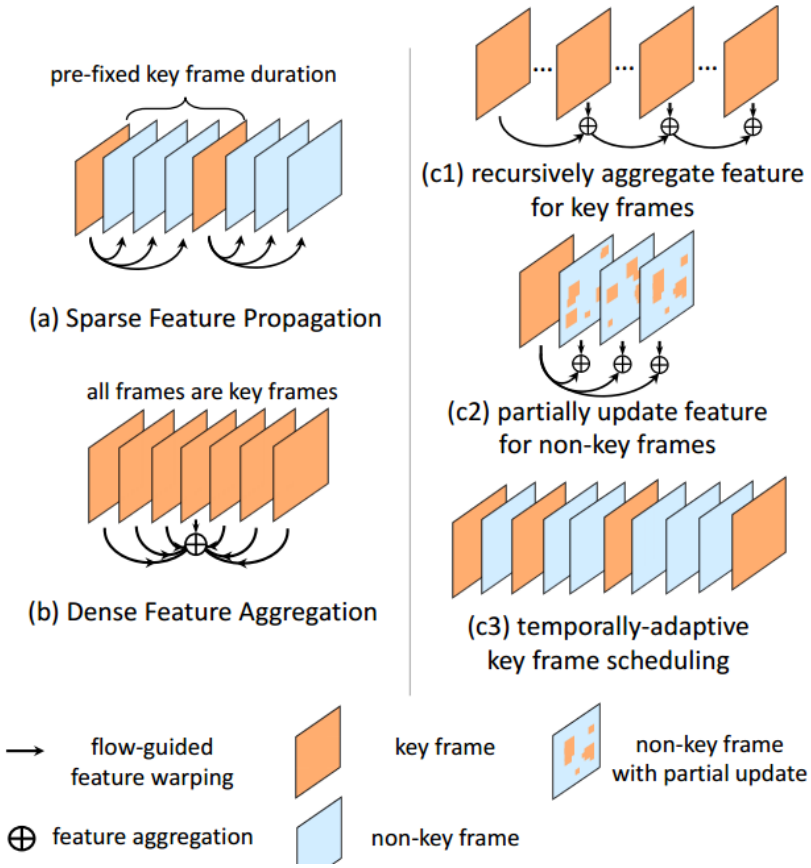


DFF

将图片分为关键帧和非关键帧，在关键帧上采用卷积提取特征，在非关键帧利用光流将关键帧的特征传播映射过来。

$$F_{k\to i} = \mathcal{W}(F_k, M_{i\to k}),$$

$$\mathcal{N}_{\text{flow}}(I_k, I_i) = M_{i\to k}$$

FGFA

所有的帧都是关键帧，在检测某一帧时，将其他帧的特征与本帧特征进行加权，增强当前帧特征。

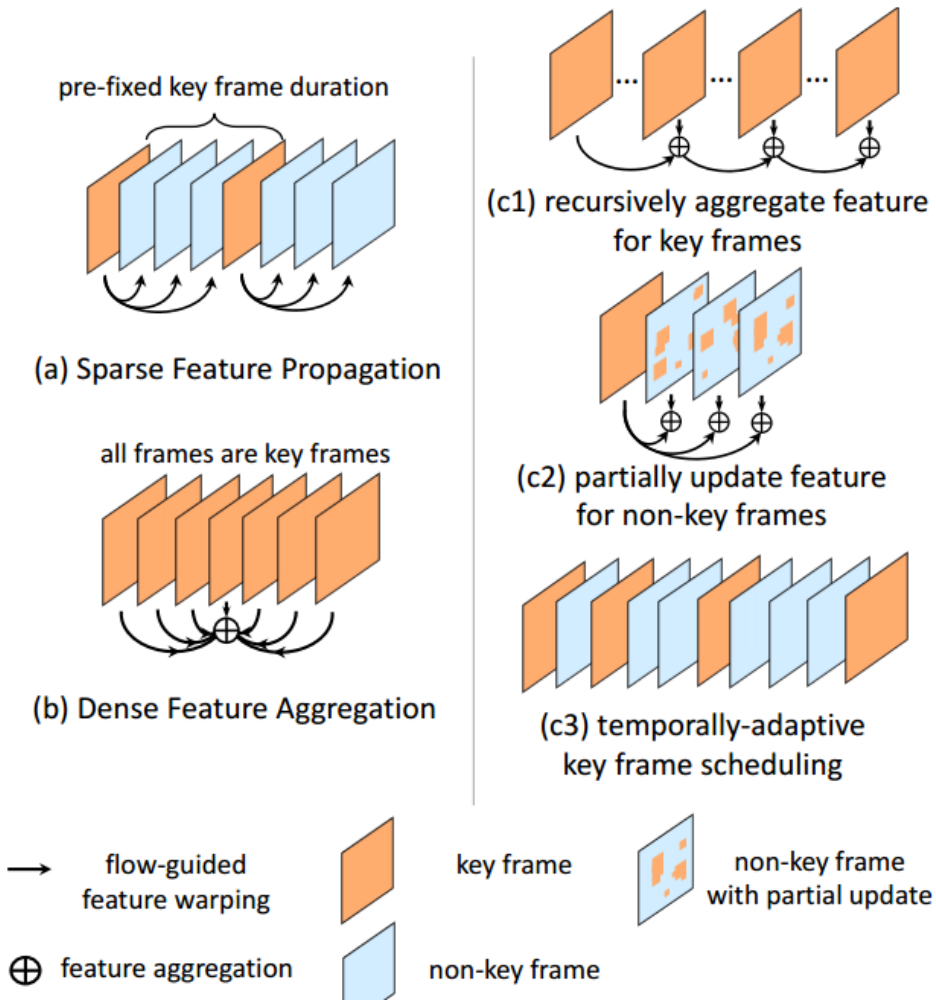
$$\bar{F}_i(p) = \sum_{k\in[i-r,i+r]} W_{k\to i}(p)\cdot F_{k\to i}(p), \forall p,$$

$$W_{k\to i}(p) = \exp\left(\frac{F^e_{k\to i}(p)\cdot F^e_i(p)}{|F^e_{k\to i}(p)|\cdot|F^e_i(p)|}\right), \forall p.$$

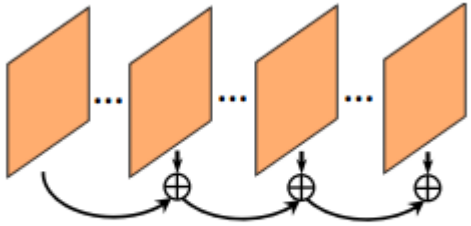
$$\sum_{k\in[i-r,i+r]} W_{k\to i}(p) = 1.$$

Method

基于之前的工作（DFF，FGFA），提出三种不同的方法来提高检测的速度和精度。



Sparsely Recursive Feature Aggregation



(c1) recursively aggregate feature for key frames

之前FGFA的特征聚合是很稠密的，所以计算量很大，这里他提出对于**key frame**而言，不做稠密的聚合而是做那种递归式地从前到后进行，这样大大减少了计算量，相当于两帧之间做聚合，后面的帧会考虑前面所有帧的**历史信息**。

$$\bar{F}_{k'} = W_{k\to k'} \odot \bar{F}_{k\to k'} + W_{k'\to k'} \odot F_{k'},$$

$$\bar{F}_{k\to k'} = \mathcal{W}(\bar{F}_k, M_{k'\to k})$$

$$W_{k\to k'}(p) + W_{k'\to k'}(p) = 1$$

Spatially-adaptive Partial Feature Updating

DFF在非关键帧上直接使用关键帧的映射，但无法保证映射的特征好坏。partially update feature代表对于不同区域，采用不同的策略，warp或者做cnn 提取特征，那么如何确定那个点是用warp还是用cnn呢，作者在光流网络上加了一个分支，输出是一个mask（记为Q）

$Q_{k\to i}$, together with motion field $M_{i\to k}$, as

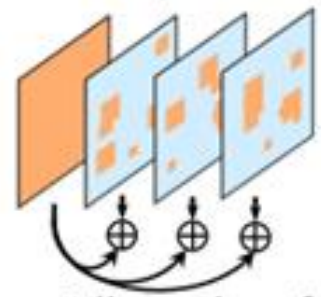
$$\{M_{i\to k}, Q_{k\to i}\} = \mathcal{N}_{\text{flow}}(I_k, I_i). \tag{5}$$

If $Q_{k\to i}(p) \le \tau$, the propagated feature $F_{k\to i}(p)$ is inconsistent with the real feature $F_i(p)$. That is to say, $F_{k\to i}(p)$ is a bad approximation, which suggests updating with real feature $F_i(p)$.

We consider a partial feature updating for non-key frames. Feature at frame i is updated by

$$\hat{F}_i = U_{k\to i} \odot \mathcal{N}_{\text{feat}}(I_i) + (1 - U_{k\to i}) \odot F_{k\to i}, \tag{6}$$

where the updating mask $U_{k\to i}(p) = 1$ if $Q_{k\to i}(p) \le \tau$, and $U_{k\to i}(p) = 0$, otherwise. In our implementation, we



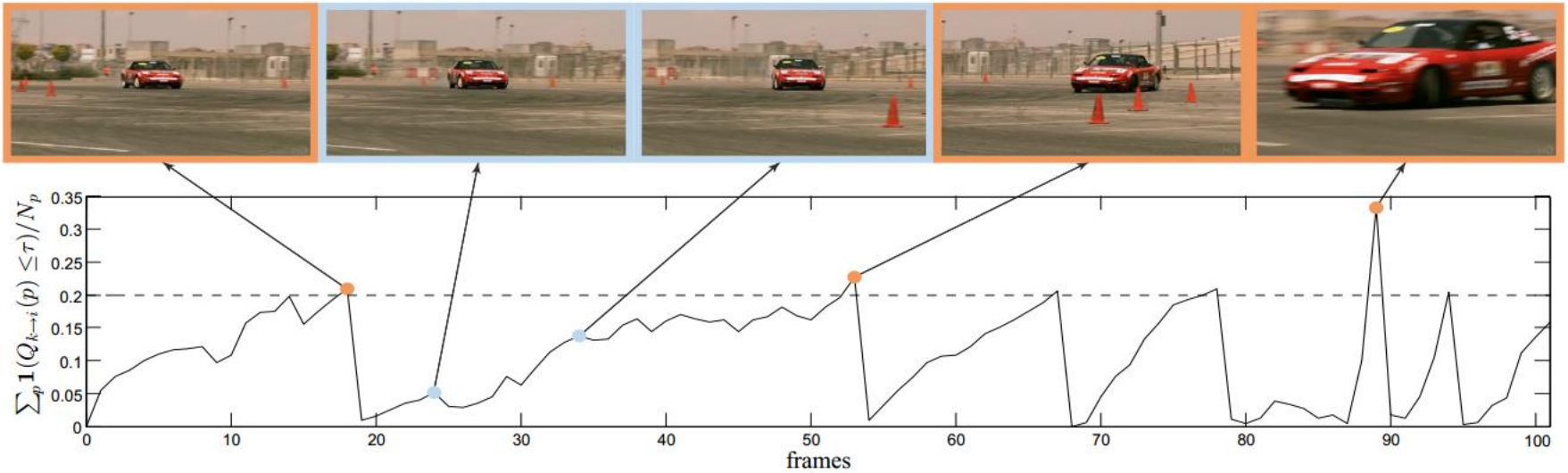
(c2) partially update feature for non-key frames

$$\bar{F}_i = W_{k\to i} \odot \bar{F}_{k\to i} + W_{i\to i} \odot \hat{F}_i,$$

Temporally-adaptive Key Frame Scheduling

固定的选取关键帧可能会影响检测精度

$$is_key(Q_{k\to i}) = [\frac{1}{N_p}\sum_p \mathbf{1}(Q_{k\to i}(p) \le \tau)] > \gamma$$



Experiments

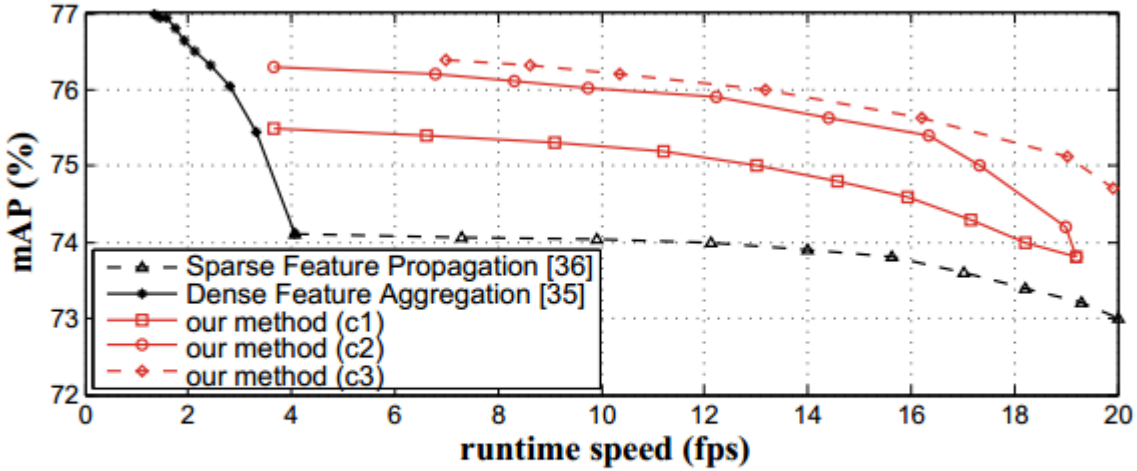


Figure 3. Speed-accuracy trade-off curves for methods in Table 1.

method	feature network	mAP (%)	runtime (fps) (TitanX/K40)
Ours	ResNet-101+DCN	**78.6** 77.8	13.0 / 8.6 22.9 / 15.2
TPN [19]	GoogLeNet	68.4	2.1 / -
D&T [8]	ResNet-101	75.8	7.8 / -
ImageNet VID 2017 winner [6]	ResNet-101	76.8	- / 15.4

Table 2. Comparison with state-of-the-art methods.